

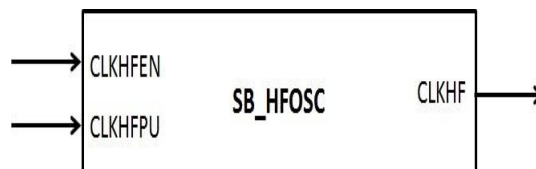
iCE5LP (iCE40 Ultra) Hard Macros

This section describes the following dedicated hard macro primitives available in iCE5LP (iCE40 Ultra) devices.

SB_HFOSC
SB_LFOSC
SB_LED_DRV_CUR
SB_RGB_DRV
SB_IR_DRV
SB_IO_OD
SB_I2C
SB_SPI
SB_MAC16

SB_HFOSC

SB_HFOSC primitive generates 48MHz nominal clock frequency within +/- 10% variation, with user programmable divider value of 1, 2, 4, and 8. The HFOSC can drive either the global clock network or fabric routes directly based on the clock network selection.



Ports

SB_HFOSC Ports		
Signal Name	Direction	Description
CLKHFPU	Input	Power up the HFOSC circuit. After power up, oscillator output will be stable after 100us. Active High.
CLKHFEN	Input	Enable the clock output. Enable should be low for the 100us power up period. Active High.
CLKHF	Output	HF Oscillator output.

Parameters

Parameter	Parameter Values.	Description.
CLKHF_DIV	"0b00"	Sets 48MHz HFOSC output.
	"0b01"	Sets 24MHz HFOSC output.
	"0b10"	Sets 12MHz HFOSC output.
	"0b11"	Sets 6MHz HFOSC output.

Default Signal Values

The iCEcube2 software assigns the following signal value to unconnected port:

Input CLKHFEN: Logic "0"

Input CLKHFPU: Logic "0"

Clock Network Selection

By default the oscillator use one of the dedicated clock networks in the device to drive the elements. The user may configure the oscillator to use the fabric routes instead of global clock network using the synthesis attribute.

Synthesis Attributes

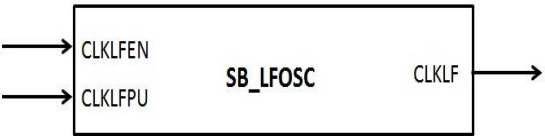
```
/* synthesis ROUTE_THROUGH_FABRIC = <value> */  
  
Value:  
0: Use dedicated clock network. Default option.  
1: Use fabric routes.
```

Verilog Instantiation

```
SB_HFOSC OSCInst0 (  
  .CLKHFEN(ENCLKHF),  
  .CLKHFPU(CLKHF_POWERUP),  
  .CLKHF(CLKHF)  
) /* synthesis ROUTE_THROUGH_FABRIC= [0|1] */;  
  
defparam OSCInst0.CLKHF_DIV = "0b00";
```

SB_LFOSC

SB_LFOSC primitive generates 10 KHz nominal clock frequency within +/- 10% variation. There is no divider on the LFOSC.
The LFOSC can drive either the global clock network or fabric routes directly based on the clock network selection



Ports

SB_LFOSC Ports		
Signal Name	Direction	Description
CLKLFPU	Input	Power up the LFOSC circuit. After power up, oscillator output will be stable after 100us. Active High.
CLKLFEN	Input	Enable the clock output. Enable should be low for the 100us power up period. Active High.
CLKLF	Output	LF Oscillator output.

Default Signal Values

The iCEcube2 software assigns the following signal value to unconnected port:
Input CLKLFEN: Logic “0”
Input CLKLFPU: Logic “0”

Clock Network Selection

By default the oscillator use one of the dedicated clock networks in the device to drive the elements. The user may configure the oscillator to use the fabric routes instead of global clock network using the synthesis attribute.

Synthesis Attributes

```
/* synthesis ROUTE_THROUGH_FABRIC = <value> */
```

Value:

0: Use dedicated clock network. Default option.

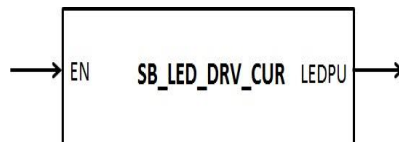
1: Use fabric routes.

Verilog Instantiation

```
SB_LFOSC OSCInst1 (  
    .CLKLFEN(ENCLKLF),  
    .CLKLFPU(CLKLF_POWERUP),  
    .CLKLF(CLKLF)  
) /* synthesis ROUTE_THROUGH_FABRIC= [0|1] */;
```

SB_LED_DRV_CUR

SB_LED_DRV_CUR primitive generates the constant reference current required to power up the SB_RGB_DRV and SB_IR_DRV primitives.



Ports

SB_LED_DRV_CUR Ports		
Signal Name	Direction	Description
EN	Input	Enable to generate constant current source for SB_RGB_DRV and SB_IR_DRV primitives. After Enable, reference current value will be stable after 100us. Active High.
LEDPU	Output	Power up signal for SB_RGB_DRV and SB_IR_DRV primitives. This port should be connected only to RGBPU/IRPU pins of SB_RGB_DRV and SB_IR_DRV primitives.

Default Signal Values

The iCEcube2 software assigns the following signal value to unconnected port:

Input EN : Logic "0"

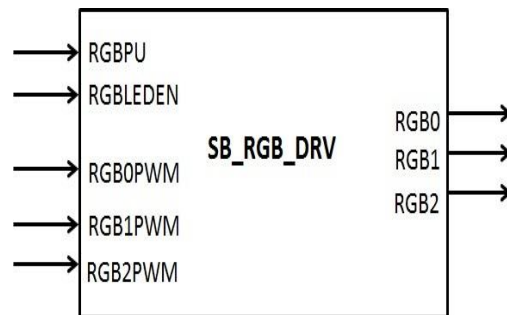
Verilog Instantiation

```
SB_LED_DRV_CUR    LED_CUR_inst
(
    .EN(enable_led_current),
    .LEDPU(led_power_up)
);
```

SB_RGB_DRV

SB_RGB_DRV primitive contains 3 dedicated open drain I/O pins for RGB LED outputs. Each of the RGB LED output is bonded out together with an SB_IO_OD primitive to the package pin. User can either use SB_RGB_DRV primitive or the SB_IO_OD primitive to drive the package pin, but not both.

The primitive allows configuration of each of the 3 RGB LED outputs individually. When the RGBx_CURRENT parameter of RGBx output is set to "0b000000", then SB_IO_OD can be used to drive the package pin.



Ports

SB_RGB_DRV Ports		
Signal Name	Direction	Description
RGBLEDEN	Input	Enable the SB_RGB_DRV primitive. Active High.
RGBPU	Input	Power Up Signal. Connect to LEDPU port of SB_LED_DRV_CUR primitive.
RGB0PWM	Input	Input data to drive RGB0 LED pin. This input is usually driven from the SB_LEDD_IP.
RGB1PWM	Input	Input data to drive RGB1 LED pin. This input is usually driven from the SB_LEDD_IP.
RGB2PWM	Input	Input data to drive RGB2 LED pin. This input is usually driven from the SB_LEDD_IP.
RGB0	Output	RGB0 LED output.
RGB1	Output	RGB1 LED output.
RGB2	Output	RGB2 LED output.

Default Signal Values

The iCEcube2 software assigns the following signal value to unconnected port:

Input RGBLEDEN : Logic "0"
Input RGB0PWM : Logic "0"
Input RGB1PWM : Logic "0"

Input RGB2PWM : Logic "0"
Input RGBPU : Logic "0"

Parameters

The SB_RGB_DRV primitive contains the following parameter and their default values:

Parameter RGB0_CURRENT = "0b000000";
Parameter RGB1_CURRENT = "0b000000";
Parameter RGB2_CURRENT = "0b000000";

Parameter values:

"0b000000" = 0mA. // Set this value to use the associated SB_IO_OD instance at RGB
// LED location.

"0b000001" = 4mA.

"0b000011" = 8mA.

"0b000111" = 12mA.

"0b001111" = 16mA.

"0b011111" = 20mA.

"0b111111" = 24mA.

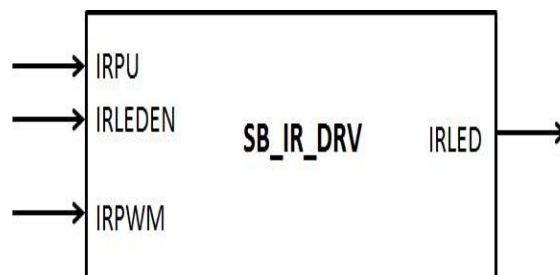
Verilog Instantiation

```
SB_RGB_DRV RGB_DRIVER (  
    .RGBLEDEN(ENABLE_LED),  
    .RGB0PWM(RGB0),  
    .RGB1PWM(RGB1),  
    .RGB2PWM(RGB2),  
    .RGBPU(led_power_up),  
    .RGB0(LED0),  
    .RGB1(LED1),  
    .RGB2(LED2)  
);  
  
defparam RGB_DRIVER.RGB0_CURRENT = "0b111111";  
defparam RGB_DRIVER.RGB1_CURRENT = "0b111111";  
defparam RGB_DRIVER.RGB2_CURRENT = "0b111111";
```

SB_IR_DRV

SB_IR_DRV primitive contains a single dedicated open drain I/O pin for IRLED output. The IRLED output is bonded out together with an SB_IO_OD primitive to the package pin. User can either use SB_IR_DRV primitive or the SB_IO_OD primitive to drive the package pin, but not both.

When the IR_CURRENT parameter is set to "0b0000000000", then SB_IO_OD can be used to drive the package pin.



Ports

SB_IR_DRV Ports		
Signal Name	Direction	Description
IRLEDEN	Input	Enable the SB_IR_DRV primitive. Active High.
IRPU	Input	Power Up Signal. Connect to LEDPU port of SB_LED_DRV_CUR primitive.
IRPWM	Input	PWM Input data to drive IRLED pin.
IRLED	Output	IR LED output.

Default Signal Values

The iCEcube2 software assigns the following signal value to unconnected port:

Input IRLEDEN : Logic "0"
Input IRPWM : Logic "0"

Parameter

The SB_IR_DRV primitive contains the following parameter and their default values:

Parameter IR_CURRENT = "0b0000000000";

Parameter Values:

```
"0b0000000000"; = 0mA. // Set this value to use the associated SB_IO_OD instance at  
                        // IR LED location.  
"0b0000000001"; = 50mA  
"0b0000000011"; = 100mA  
"0b0000000111"; = 150mA  
"0b0000001111"; = 200mA  
"0b0000011111"; = 250mA  
"0b0000111111"; = 300mA  
"0b0001111111"; = 350mA  
"0b0011111111"; = 400mA  
"0b0111111111"; = 450mA  
"0b1111111111"; = 500mA
```

Verilog Instantiation

```
SB_IR_DRV  IRDRVinst (  
    .IRLEDEN(ENABLE_IRLED),  
    .IRPWM(IR_INPUT),  
    .IRPU(led_power_up),  
    .IRLED(IR_LED)  
);  
defparam IRDRVinst.IR_CURRENT = "0b1111111111";
```

SB_RGB_IP

SB_RGB_IP primitive generates the 3 RGB PWM outputs, to be connected to the LED drivers.



Ports

SB_RGB_IP Ports		
Signal Name	Direction	Description
CLK	Input	Clock Input.
RST	Input	Asynchronous Reset Input.
PARAMSOK	Input	Enable signal to sample the rgb data.
RGBCOLOR[3:0]	Input	4 bit rgb color data input
BRIGHTNESS[3:0]	Input	4 bit rgb brightness data input
BREATHRAMP[3:0]	Input	4 bit breathramp data input
BLINKRATE[3:0]	Input	4 bit blinkrate data input
REDPWM	Output	RED PWM output.
GREENPWM	Output	GREEN PWM output.
BLUEPWM	Output	BLUE PWM output.

Default Signal Values

The iCEcube2 software assigns the logic “0” value to all unconnected input ports.

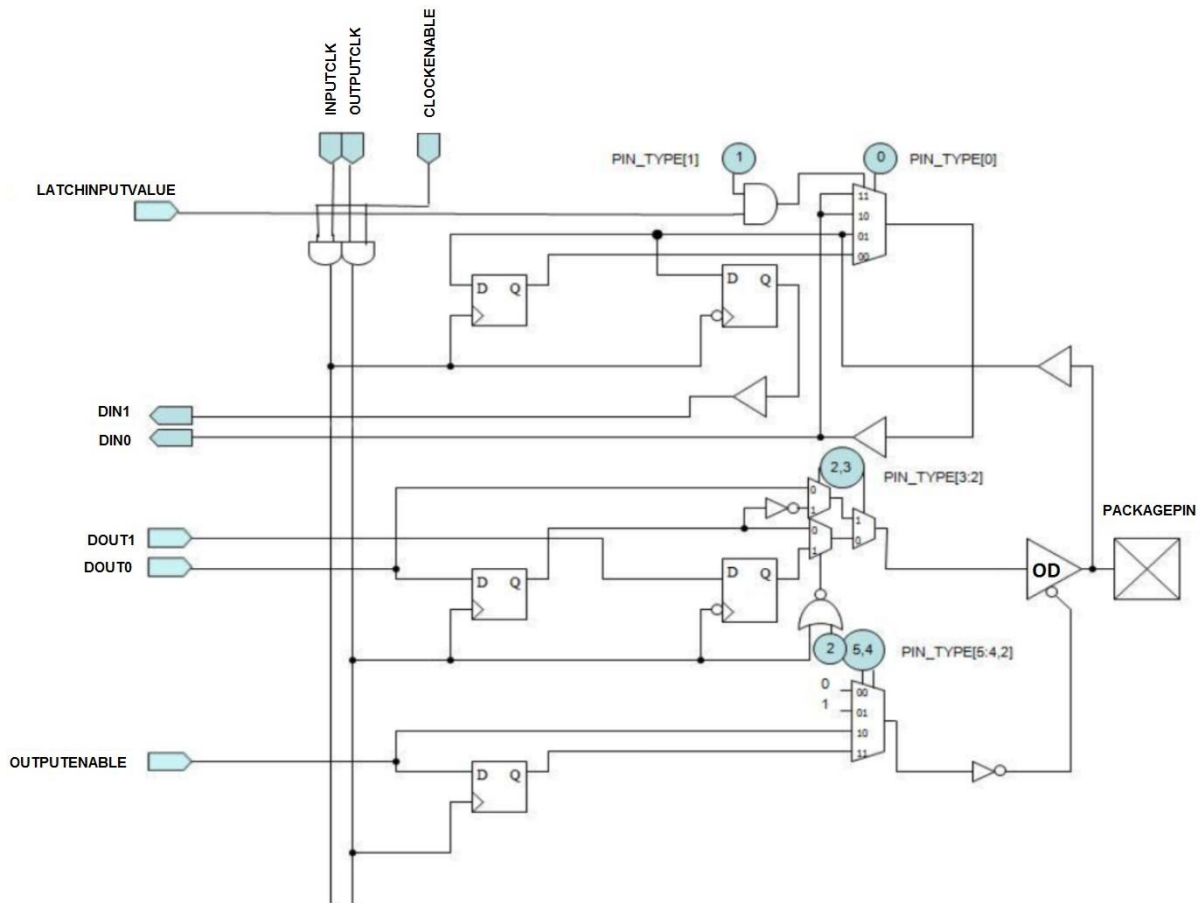
Verilog Instantiation

```

SB_RGB_IP RGBPWMIP_inst(
    .CLK (CLK),
    .RST (RST),                      // Async rst
    .PARAMSOK (PARAMSOK),
    .RGBCOLOR (RGBCOLOR),
    .BRIGHTNESS (BRIGHTNESS),
    .BREATHRAMP (BREATHRAMP),
    .BLINKRATE (BLINKRATE),
    .REDPWM (REDPWM),
    .GREENPWM (GREENPWM),
    .BLUEPWM (BLUEPWM)
);
  
```

SB_IO_OD

The SB_IO_OD is the open drain IO primitive. When the Tristate output is enabled, the IO pulls down the package pin signal to zero. The following figure and Verilog template illustrate the complete user accessible logic diagram, and its Verilog instantiation.



Ports

SB_IO_OD Ports		
Signal Name	Direction	Description
PACKAGEPIN	Bidirectional	Bidirectional IO pin.
LATCHINPUTVALUE	Input	Latches/Holds the input data.
CLOCKENABLE	Input	Clock enable signal.
INPUTCLK	Input	Clock for the input registers.
OUTPUTCLK	Input	Clock for the output registers.
OUTPUTENABLE	Input	Enable Tristate output. Active high.
DOUT0	Input	Data to package pin.
DOUT1	Input	Data to package pin.
DIN0	Output	Data from package pin.
DIN1	Output	Data from package pin.

Default Signal Values

The iCEcube2 software assigns the logic “0” value to all unconnected input ports except for CLOCKENABLE.

Note that explicitly connecting logic “1” value to port CLOCKENABLE will result in a non-optimal implementation, since extra LUT will be used to generate the Logic “1”. If the user’s intention is to keep the Input and Output registers always enabled, it is recommended that port CLOCKENABLE to be left unconnected.

Parameter Values

```
Parameter PIN_TYPE = 6'b000000;  
                                // See Input and Output Pin Function Tables in SB_IO.  
                                // Default value of PIN_TYPE = 6'b000000  
Parameter NEG_TRIGGER = 1'b0;  
                                // Specify the polarity of all FFs in the I/O to be falling edge when  
                                NEG_TRIGGER = 1. Default is 1'b0, rising edge.
```

Input and Output Pin Function Tables

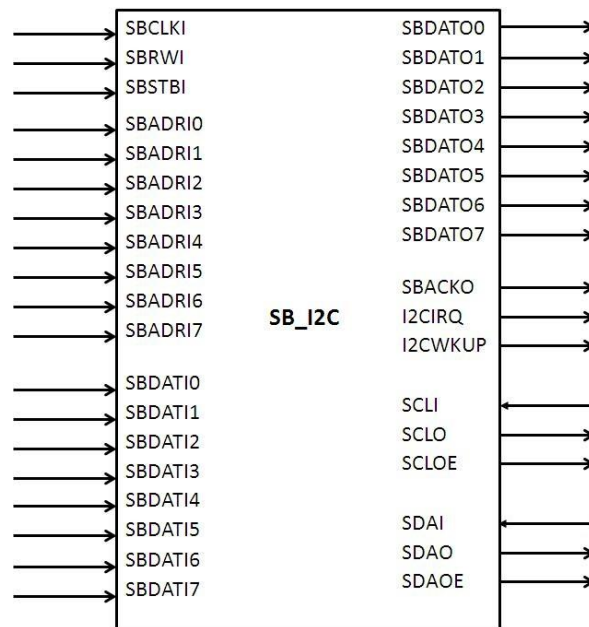
Refer SB_IO Input and Output Pin Functional Table for the PIN_TYPE settings. Some of the output pin configurations are not applicable for SB_IO_OD primitive.

Verilog Instantiation

```
SB_IO_OD    OpenDrainInst0  
(  
    .PACKAGEPIN (PackagePin),           // User's Pin signal name  
    .LATCHINPUTVALUE (latchinputvalue), // Latches/holds the Input value  
    .CLOCKENABLE (clockenable),          // Clock Enable common to input and  
                                         // output clock  
    .INPUTCLK (inputclk),                // Clock for the input registers  
    .OUTPUTCLK (outputclk),              // Clock for the output registers  
    .OUTPUTENABLE (outputenable),        // Output Pin Tristate/Enable  
                                         // control  
    .DOUT0 (dout0),                      // Data 0 - out to Pin/Rising clk  
                                         // edge  
    .DOUT1 (dout1),                      // Data 1 - out to Pin/Falling clk  
                                         // edge  
    .DIN0 (din0),                        // Data 0 - Pin input/Rising clk  
                                         // edge  
    .DIN1 (din1),                        // Data 1 - Pin input/Falling clk  
                                         // edge  
);  
  
defparam OpenDrainInst0.PIN_TYPE = 6'b000000;  
defparam OpenDrainInst0.NEG_TRIGGER = 1'b0;
```

SB_I2C

iCE5LP device supports two I2C hard IP primitives , located at upper left corner and upper right corner of the chip.



Ports

The port interface is similar as iCE40LM SB_I2C primitive. Refer Page 114.

Parameters

The parameters are same as ICE40LM SB_I2C primitive.

Synthesis Attribute

I2C_CLK_DIVIDER

Synthesis attribute "I2C_CLK_DIVIDER" is used by PNR and STA tools for optimization and deriving the appropriate clock frequency at SCLO output with respect to the SBCLKI input clock frequency.

```
/* synthesis I2C_CLK_DIVIDER= Divide Value */
```

Divide Value: 0, 1, 2, 3 ... 1023. Default is 0.

SDA_INPUT_DELAYED

SDA_INPUT_DELAYED attribute is used to add 50ns additional delay to the SDAI signal.

```
/* synthesis SDA_INPUT_DELAYED= value */
```

Value: